

Question 1

(a) Given $r(x, y) = \sqrt{x^2 + y^2} = (x^2 + y^2)^{\frac{1}{2}}$

$$\begin{aligned}
 r_x &= \frac{1}{2}(x^2 + y^2)^{-\frac{1}{2}} \cdot 2x \\
 &= x(x^2 + y^2)^{-\frac{1}{2}} \\
 &= \frac{x}{r} \\
 r_{xx} &= \frac{1}{r} - \frac{1}{2}x(x^2 + y^2)^{-\frac{3}{2}} \cdot 2x \\
 &= \frac{1}{r} - x^2(x^2 + y^2)^{-\frac{3}{2}} \\
 &= \frac{1}{r} - \frac{x^2}{r^3} \\
 &= \frac{r^2 - x^2}{r^3} \\
 &= \frac{y^2}{r^3} \\
 r_y &= \frac{y}{r} \\
 r_{yy} &= \frac{x^2}{r^3}
 \end{aligned}$$

(b) Given $u(x, y) = f(r(x, y))$

$$\nabla^2 u = u_{xx} + u_{yy}$$

Find f_{xx} and f_{yy}

$$\begin{aligned}
 f_x &= f'(r)r_x \\
 f_{xx} &= f''(r)r_x + f'(r)r_{xx} \\
 &= f''(r)\frac{x}{r} + f'(r)\frac{y^2}{r^3} \\
 &= \frac{1}{r} \left(f''(r)x + f'(r)\frac{y^2}{x^2 + y^2} \right) \\
 f_{yy} &= \frac{1}{r} \left(f''(r)y + f'(r)\frac{x^2}{x^2 + y^2} \right)
 \end{aligned}$$

So we have

$$\begin{aligned}\nabla^2 u &= \frac{1}{r} \left(f''(r)x + f'(r)\frac{y^2}{x^2 + y^2} + f''(r)y + f'(r)\frac{x^2}{x^2 + y^2} \right) \\ &= \frac{1}{r} (f''(r)(x + y) + f'(r)) \\ &= f''(r)\frac{x + y}{r} + \frac{1}{r}f'(r)\end{aligned}$$

(c) So we have the ODE $f''(r) + \frac{1}{r}f'(r) = 0$

$$\begin{aligned}f''(r) &= -\frac{1}{r}f'(r) \\ \frac{d}{dr}f'(r) &= -\frac{1}{r}f'(r)\end{aligned}$$

Let $u = f'(r)$

$$\begin{aligned}\frac{du}{dr} &= -\frac{1}{r}u \\ \frac{1}{u}du &= -\frac{1}{r}dr \\ \ln u &= -\ln r + c \\ u &= \frac{C}{r} \\ f'(r) &= \frac{C}{r} \\ \frac{d}{dr}f(r) &= \frac{C}{r} \\ f(r) &= C \ln |r| + B\end{aligned}$$

Question 2

- (a) Denote t as y . The wave equation becomes $u_{yy} - c^2 u_{xx} = 0$. We have the general form of a constant-coefficients PDE as

$$Au_{xx} + Bu_{xy} + Cu_{yy} + F(x, y, u, u_x, u_y) = 0$$

In this case, we have $A = -c^2, B = 0, C = 1$, so our discriminant is

$$D = 0^2 - 4 \cdot -c^2 \cdot 1 = 4c^2 > 0$$

Therefore the wave equation is hyperbolic.

- (b) Given $x = \frac{1}{2}\xi + \frac{1}{2}\eta$ and $t = \frac{1}{2c}\xi - \frac{1}{2c}\eta$

$$\begin{aligned} u_\xi &= u_x x_\xi + u_t t_\xi \\ u_{\xi\xi} &= \frac{\partial}{\partial \xi}(u_x x_\xi) + \frac{\partial}{\partial \xi}(u_t t_\xi) \end{aligned}$$

Question 3

- (a) Given $u(x, t) = 2kt + x^2$ and the heat equation $u_t - ku_{xx} = 0$.

$$\begin{aligned} u_t &= 2k \\ u_x &= 2x \\ u_{xx} &= 2 \\ 2k - k \cdot 2 &= 0 \\ 0 &= 0 \quad \checkmark \end{aligned}$$

Try $u(x - y, t) = 2kt + x^2 - 2xy + y^2$

$$\begin{aligned} u_t &= 2k \\ u_x &= 2x - 2y \\ u_{xx} &= 2 \\ 2k - k \cdot 2 &= 0 \\ 0 &= 0 \quad \checkmark \end{aligned}$$

As $x \rightarrow \pm\infty$, u_x does not approach zero, therefore it does not satisfy the decay condition.

(b) Given $v(x, t) = \int_0^1 u(x - y, t) y^2 dy$

$$\begin{aligned}
 v(x, t) &= \int_0^1 (2kt + x^2 - 2xy + y^2) y^2 dy \\
 &= \int_0^1 2kty^2 + x^2 y^2 - 2xy^3 + y^4 dy \\
 &= \frac{2}{3} kty^3 + \frac{1}{3} x^2 y^3 - \frac{1}{2} xy^4 + \frac{1}{5} y^5 \Big|_0^1 \\
 &= \frac{2}{3} kt + \frac{1}{3} x^2 - \frac{1}{2} x + \frac{1}{5} \\
 v_t &= \frac{2}{3} k \\
 v_x &= \frac{2}{3} x - \frac{1}{2} \\
 v_{xx} &= \frac{2}{3} \\
 \frac{2}{3} k - k \cdot \frac{2}{3} &= 0 \\
 0 &= 0 \quad \checkmark
 \end{aligned}$$

$v(x, t)$ does solve the heat equation.

(c)

$$\begin{aligned}
 v_t &= \int_0^1 u_t(x - y, t) y^2 dy \\
 \frac{2}{3} k &= \int_0^1 (2k) y^2 dy \\
 &= \frac{2}{3} ky^3 \Big|_0^1 \\
 &= \frac{2}{3} k \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 v_{xx} &= \int_0^1 u_{xx}(x-y, t) y^2 dy \\
 \frac{2}{3} &= \int_0^1 2y^2 dy \\
 &= \frac{2}{3} y^3 \Big|_0^1 \\
 &= \frac{2}{3} \quad \checkmark
 \end{aligned}$$

Question 4

(a) Let $w(x, t) = u(\sqrt{ax}, at)$

$$w_t =$$

(b) Suppose $u(x, t) = f\left(\frac{x}{\sqrt{4kt}}\right)$

$$\begin{aligned}
 u_t &= f'(z) \cdot -\frac{1}{2} \frac{x}{\sqrt{4kt^3}} \\
 &= f'(z) \cdot -\frac{1}{2t} z \\
 u_x &= f'(z) \cdot \frac{1}{\sqrt{4kt}} \\
 u_{xx} &= f''(z) \cdot \frac{1}{4kt}
 \end{aligned}$$

Plugging this into the heat equation, we get

$$\begin{aligned}
 f'(z) \cdot -\frac{1}{2t} z - k f''(z) \cdot \frac{1}{4kt} &= 0 \\
 -\frac{1}{2t} f'(z) - \frac{1}{4t} f''(z) &= 0 \\
 f''(z) + 2f'(z) &= 0
 \end{aligned}$$

Question 5

Given $p(x, t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-\mu)^2}{4kt}}$

$$p_t = \frac{-1}{2t\sqrt{4\pi kt}} e^{-\frac{(x-\mu)^2}{4kt}} + \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-\mu)^2}{4kt}} \cdot \frac{(x-\mu)^2}{4kt^2}$$

$$\begin{aligned} p_x &= \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-\mu)^2}{4kt}} \cdot \frac{-2(x-\mu)}{4kt} \\ &= \frac{-(x-\mu)}{2kt\sqrt{4\pi kt}} e^{-\frac{(x-\mu)^2}{4kt}} \end{aligned}$$

$$p_{xx} =$$